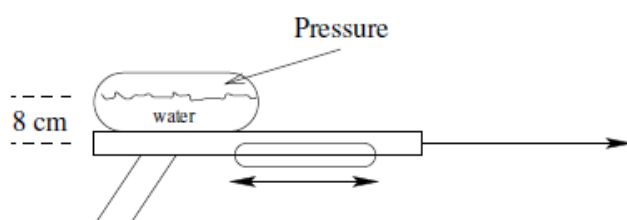


The Super Soaker water gun was a revolution in water gun technology. The pump handle is moved back and forth building up significant pressure in the water reservoir. The water is released by pulling a trigger and shoots for a significant distance. Assume that one can make an *absolute pressure* of 8 atmospheres in the reservoir. Find all the following:

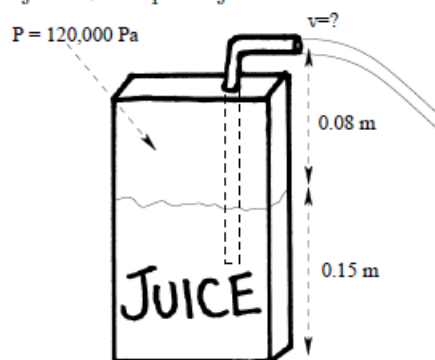


- What is the velocity at which the water leaves the gun?
- If the water exits the gun through a hole with radius 1mm, what is the volume rate of flow ($\frac{m^3}{s}$)?
- If the water gun is fired horizontally and held 1.2 meters above the ground, where does the water hit the ground?

NOTE: The reservoir is a “tank” of water in which the water moves with negligible velocity..

PROBLEMS-Fluids

A child has been drinking from a juice box and decides to play a game. He blows bubbles into the juice box pressurizing it to an absolute pressure of 120,000 Pa. To his surprise, when he takes the straw out of his mouth, the juice box squirts juice all over him!

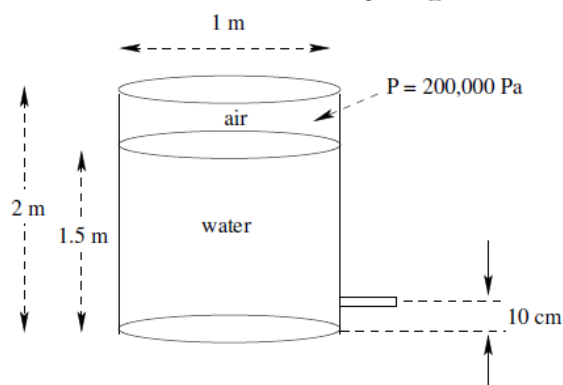


NOTES:

- Use $P = 100,000 \text{ Pa}$ as atmospheric pressure.
- Approximate the density of the juice to be the same as water, $\rho = 1000 \frac{\text{kg}}{\text{m}^3}$.
- Approximate the velocity of the juice to be zero while it is still in the box.

- (a) (7 pts) Calculate the velocity of the juice as it leaves the straw.
- (b) (7 pts) If the radius of the straw is $r = 0.001 \text{ m}$, calculate the volume rate of flow $\frac{\text{m}^3}{\text{s}}$ of the juice flowing out of the straw.
- (c) (6 pts) Explain in one to two sentences why the juice box stops squirting before all the juice is expended.

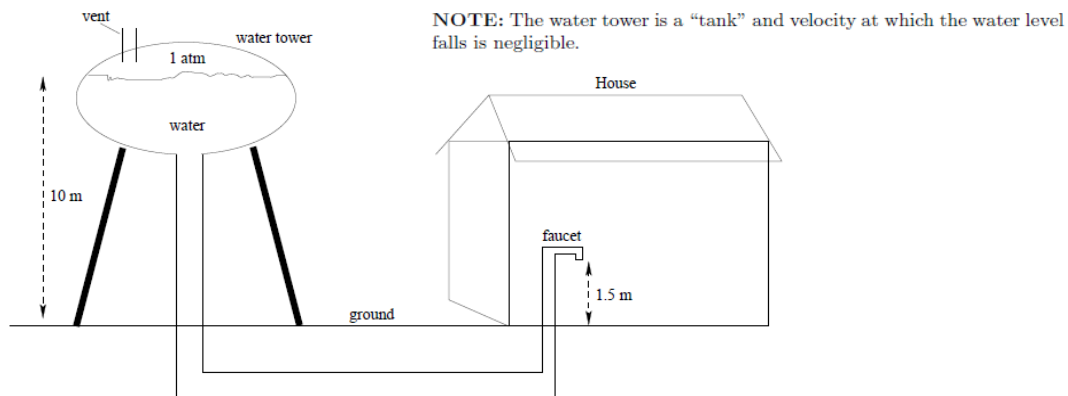
Can't Drain the Water The figure below shows a water tank that is *closed* at the top. The 2 meter tall tank contains 1.5 meters of water and the rest is air at a *gauge* pressure of 200,000 Pa. You attempt to drain the water tank through a faucet (or tap) located 10 cm above the base of the tank. The faucet has a circular opening with a diameter of 1 cm.



- Determine the speed of the water at the exit of the faucet at the moment when the faucet is first opened.
- Determine the speed of the water at its highest point in the tank at the same moment as in the previous part of the problem.
- The water stops flowing when the water level drops to a height of 0.91 meters. What is the pressure of the air in the tank at this point? Explain using 1 or 2 short sentences why the water has stopped flowing.

PROBLEMS-Fluids

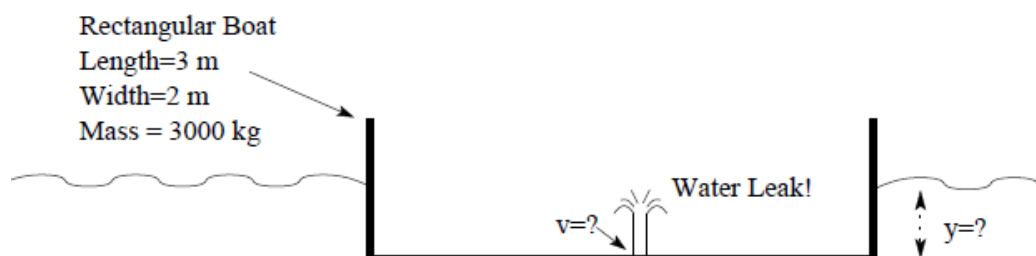
Many communities use a “water tower” to supply water under pressure to homes as in the figure below. The top of the water tower is open to the air so that the pressure above the surface of the water is 1 atmosphere. When you open the faucet in your home, water comes out. The radius of the opening at the water faucet is 0.005 meters.



- (a) (10 pts) What is the velocity of the water as it leaves the faucet?
- (b) (6 pts) What is the “Volume Rate of Flow” ($\frac{m^3}{s}$) of the water leaving the faucet?
- (c) (4 pts) Calculate the time required to fill a 1 liter bottle with water.

PROBLEMS-Fluids

You set out upon the water in a simple rectangular boat with mass $m = 3000\text{ kg}$, length $L = 3\text{ m}$, and width $w = 2\text{ m}$ as shown in the figure below.



- (a) (7 pts) Calculate the depth of the bottom of the boat, y , when it floats at equilibrium.
- (b) (7 pts) **OH NO!!!**. Your boat has a leak in the bottom. Calculate the velocity of the water coming through the hole.
- (c) (6 pts) If the hole is circular with radius $r = 0.005\text{ m}$, calculate the volume rate of flow ($\frac{\text{m}^3}{\text{s}}$) of water entering your boat.